

L2-1.2. Log Operations

1. Log Configuration Options

Log configuration is a multifaceted operation that addresses security, operational stability, regulatory compliance, and debugging needs.

Security Purposes: Logging designed to detect threats, enforce authentication, and protect system integrity and confidentiality.

Operational Purposes: Logging used to monitor system health, troubleshoot issues, and improve performance and reliability.

Legal Purposes: Logging implemented to meet regulatory, compliance, and audit requirements (e.g., GDPR, PCI DSS).

Debug Purposes: Logging used during development to identify bugs, improve visibility, and enhance application functionality.

Answer the questions below

Which of the given log purposes would be suitable to measure the cost of using a service? Operational

Which of the given log purposes would be suitable for investigating application logs for enhancement and stability? Debug

- What will you log, and for what (asset scope and logging purpose)?
 - Is additional commitment or effort required to achieve the purpose (requirements related to the purpose)?
- How much are you going to log (detail scope)?
- How much do you need to log?
- How are you going to log (collection)?
- How are you going to store collected logs?
 - Is there a standard, process, legislation, or law that you must comply with due to the data you log?
- How are you going to protect the logs?

- How are you going to analyse collected logs?
- Do you have enough resources and workforce to do logging?
- Do you have enough budget to plan, implement and maintain logging?

Answer the questions below

You are a consultant working for a growing startup. As a consultant, you participated in a log configuration planning session. The company you work for is working to get compliant to process payment information. The given question set is being discussed.

Which question's answer can be "as much as mentioned in the PCI DSS requirements."? How much do you need to log?

Let's take a closer look at exactly how the dilemma arose.

Base Requirements	Aspirations for Better Insights
• What happened? • When did it happen? ◦ With time data (if possible). • Where did it happen? ◦ Network, system, folder, path, interface. • Who/What caused it to occur? • From which log source?	• Is it possible to have more data? ◦ More details. • How sure can I be that this is true? • What is affected? • What will happen next? • Is there anything else that requires attention? • What should I do about the incident?

Answer the questions below

The session continues, and your teammates need your help; they will negotiate for logging budget and operation details. As a consultant, you must remind them of a vital point: Which requirements are non-negotiable? operational and security requirements

3. Logging Principles

Collection	• Define the logging purpose. • Collect what you will need and use. • Do not collect irrelevant data. • Avoid log noise.
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Format	• Log at the correct level and detail. • Implement a consistent log format. • Ensure that timestamps in logs are accurate and synchronised.
Archiving and Accessibility	• Define log retention policies and implement them. • Store log data and make sure the important part is available for analysis. • Create backups of stored log data and used systems.
Monitoring and Alerting	• Create alerts and notifications for important and noteworthy cases. • Focus on actionable alerts and avoid noise.
Security	• Protect logs by implementing access controls. • Implement encryption if required. • Use a dedicated log management solution.
Continuous Change	• Logging sources, types, and messages are constantly changing and being updated. ◦ Be open to continuous change. • Train your personnel.

4. Challenges

Data Volumeand Noise	• Having multiple data sources to deal with. • Differences in the log volumes created by applications. ◦ Some applications generate an insufficient amount of logs. ◦ Large-scale applicants could generate massive log volumes. • Some logs can provide non-essential data and make the identifying process difficult.
System Performanceand Collection	• Log collection can slow down the system's performance. • Systems are not always "state of the art". ◦ Some "sensitive" or "ancient" systems are impossible to touch. • Deployment and optimisation challenges. ◦ Managing system and agent version updates and synchronisation in large-scale networks is overwhelming.

Process and Archive	• Having multiple data formats to deal with it. ◦ Parsing different data sources and formats is time-consuming and error-prone. • Balancing the log retention can be challenging. ◦ Especially when dealing with many compliance regulations and standards.
Security	• Ensuring data security is a task/challenge in itself.
Analysis	• Combining, correlating, and analysing data from multiple sources to understand the context of an incident is a time-consuming process that requires significant computing resources and expertise. ◦ Achieving this in real-time is also another challenge in the same scope. ◦ Avoiding false positives/negatives is overwhelming.
Misc	• Lack of planning and roadmap. • Lack of financial resources/budget. • Lack of implementation scenarios, playbooks, and exercises. • Lack of technical skills to implement, maintain, and analyse. • Focusing on log collection instead of the analysis phase. • Ignoring human factors and potential system errors.

Mistakes"don'ts"	Best Practices"dos"
• Logging sensitive information! • Creating logs by yourself. • Having uncollected logs. • Collecting everything but not analysing. • Collecting logs without proper planning and configuration. • Having systems that lack the planned/required log configuration. • Skipping the scale, testing, and functionality analysis. • Focusing on edges and skipping the internal systems in analysis.	• Create a suitable log configuration and plan according to your systems. • Implement testing on scale, functionality, and operational stability. • Exclude logging sensitive information! • Secure your logs. • Create meaningful alerts/notifications. • Focus on having insights on actionable and impactful results. • Train your analysts and enhance their skills. • Update/maintain your operation plans and components/assets as needed.

Mistakes "don'ts"	Best Practices "dos"
• "Searching for what you want to find" and "Not investigating what you see". • Forgetting that the process takes the form of proper planning, management, and analysis.	

Answer the questions below

Your team is working on policies to decide which logs will be stored and which portion will be available for analysis. Which of the given logging principles would be implemented and improved? Archiving and Accessibility

Your team implemented a brand new API logging product. One of the team members has been tasked with collecting the logs generated by that new product. The team member reported continuous errors when transferring the logs to the review platform. In this case, which of the given difficulties occurs? Process and Archive

As a consultant, you are doing a comprehensive risk assessment and noticed that one of the development teams implemented a custom script to generate logs for an old system, which omits loggings at some phases. What you would call this? (Mistake or Practice?) Mistake